

Samuel Andryasian

Sylmar, CA (818)-668-4642 samuelandryas@gmail.com www.linkedin.com/in/samuelandryasian

OBJECTIVE

Mechanical Engineer with a background working with autonomous vehicle designs, and suspension systems seeking a full time position with a challenging firm that will allow me to utilize my skills in CAD Software to develop solutions for challenges.

EDUCATION

California State Polytechnic University

Pomona, CA

Bachelor of Science, Mechanical Engineering

Expected 2027

Relevant Coursework: Mechanical Vibrations, Material Science, Manufacturing Processes, Fluid Mechanics, Machine Design, Stress Analysis

EXPERIENCE

Newports Flavor, Engineering Intern

May 2026 - August 2026

- Developed Fusion 360 CAD concepts for an automated material handling system while accounting for equipment interfaces and facility constraints.
- Performed conveyor speed, material throughput, and component motion calculations to define operating requirements for synchronized material transfer.
- Prepared and presented an engineering feasibility analysis to support future production automation decisions.

Autonomous Systems Laboratory, Research Assistant

July 2024 - September 2025

- Designed and integrated vehicle systems from CAD through installation across 2 builds to manage reliable system integration
- Produced CAD assemblies and engineering drawings to enhance component fit, serviceability, and installation
- Developed repeatable installation and inspection procedures, reducing rework time by 30%

PROJECTS

Mechanical Design Team Member

July 2024 - Present

Autonomous Vehicle Laboratory

- Used SolidWorks, dimensional measurement, and coordinate system development to establish the vehicle reference geometry and map camera, LiDAR, GPS, and networking hardware locations, enabling the computer science team to build and successfully simulate a digital twin of the vehicle.
- Used SolidWorks, mount design, rapid prototyping, fastener selection, and tolerance and fit checks to design and fabricate an aligned depth and thermal camera mount, enabling the computer vision team to capture road imagery for image segmentation and path planning.
- Used Fusion360, Mastercam, CNC machining, setup planning, and technical drawings to design and manufacture a mounting bracket and pedal connector that integrated a linear actuator with the braking system, successfully bringing the vehicle to a stop during three test cycles.

Mechanical Designer

April 2025 - June 2025

Three Degree of Freedom Robotic Arm

- Designed the robotic arm and performed force, moment, and motor torque calculations to determine the torque required at each joint for lifting and rotating the can.
- Built and assembled the arm structure, mechanical joints, gripper, and servo motors into a functional electromechanical prototype.
- The completed Arduino controlled prototype successfully picked up a can, transferred it 90 degrees to the side, rotated it 180 degrees, and placed it at the target location in approximately seven seconds.

SKILLS

Solidworks, Mastercam, CNC machining, setup planning, technical drawings, rapid prototyping, mount design, fastener selection, tolerance and fit checks, mechanical integration, soldering

ORGANIZATIONS

Autonomous Vehicle Laboratory, Cal Poly Pomona | Vice President | 2024 to 2025

American Society of Mechanical Engineers | Member | September 2026 to Present